ψ-Field Energy Harvesting and Storage Systems

Inventor: Gary Thomas Alcock

October 11, 2025

Cross-Reference to Related Applications

Pursuant to 35 U.S.C. §119(e), this application claims the benefit of the following U.S. provisional applications, the entirety of each being incorporated herein by reference:

- U.S. Provisional App. No. **63/865,279** (Confirmation No. 5785), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Gravitational and Signal Correction.*
- U.S. Provisional App. No. **63/865,283** (Confirmation No. 5905), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Energy Transmission.*
- U.S. Provisional App. No. **63/865,293** (Confirmation No. 6649), filed **Aug. 16, 2025**, entitled *Systems and Methods Using Density Field Dynamics for Information Processing, Communication, and Advanced Computing.*
- U.S. Provisional App. No. **63/868,073** (Confirmation No. 8739), filed **Aug. 21, 2025**, entitled *Methods, Systems, and Apparatus for Artificial Generation, Control, and Application of Scalar Density Fields.*
- U.S. Provisional App. No. **63/893,633** (Confirmation No. 5179), filed **Oct. 4, 2025**, entitled *Systems and Methods for Electromagnetic Coupling, Actuation, and Application of Scalar Refractive Fields.*

This application also incorporates by reference, in their entirety for all purposes, **all other provisional and non-provisional applications filed by the same inventor on or after August 15, 2025**, including any continuations, divisionals, or continuations-in-part thereof. The foregoing applications are incorporated by reference in their entirety for all purposes, including definitions, figures, and embodiments. To the extent of any inconsistency, the present disclosure controls.

Statement Regarding Federally Sponsored Research or Development

No federal funding was used in connection with the inventions described herein. No government rights apply.

Incorporation by Reference (Additional Materials)

In addition to the applications identified above, technical materials authored by the inventor concerning scalar refractive field (ψ) dynamics, cavity–atom LPI tests, navigation/timing corrections, energy coupling, communication architectures, and quantum control are incorporated

by reference. All essential material for the claimed inventions is included explicitly herein; incorporated materials are provided for background and alternative implementations.

Definitions; Conventions

As used herein, "comprising" is open-ended and does not exclude additional elements or steps; "including" and "having" are used in a like manner. Numerical ranges include all sub-ranges and individual values unless otherwise specified. Singular terms include the plural and vice versa unless context dictates otherwise. Functional descriptions are supported by the corresponding structures, algorithms, and examples disclosed herein; no limitation to "means-plus-function" under 35 U.S.C. §112(f) is intended unless expressly recited. Throughout, the scalar refractive field is denoted ψ, with optical index $n = e^{\psi}$ where applicable.

No Admission

Any discussion of references, techniques, or prior work in the Background is provided solely for context and is not an admission that any such material constitutes prior art, is known, conventional, or required.

Abstract

Disclosed are systems and methods for harvesting, converting, and storing energy from controlled scalar refractive fields (ψ). Energy is extracted from field gradients, oscillations, or interference patterns of the ψ, converting refractive-potential energy into usable electrical, thermal, or mechanical output. Devices include resonant cavities, metamaterial converters, and solid-state storage media that operate on refractive coupling rather than charge displacement, enabling high-density, low-loss energy systems.

Field of the Invention

This invention relates to energy harvesting, storage, and conversion technologies. More particularly, it concerns extraction and containment of energy from a scalar refractive field (ψ) and conversion of such energy into practical electrical or mechanical power using resonant or metamaterial coupling architectures.

Background

Conventional energy systems rely on electromagnetic induction or chemical reactions. No prior art utilizes direct coupling to a refractive potential gradient to derive usable work. Within the Density Field Dynamics framework, spatial or temporal variations in ψ correspond to real energy densities $u = \frac{c^4}{8\pi G}|\nabla\psi|^2$. Harnessing this density enables reversible energy exchange between refractive and electromagnetic domains, yielding scalable energy-harvesting mechanisms.

Summary

A converter system comprises: (1) a cavity or metamaterial structure configured to sustain oscillations in ψ; (2) coupling elements that transduce the oscillatory refractive potential into electrical current or mechanical motion; and (3) a storage module, such as dielectric or superconducting resonators, to accumulate and release energy as required. Active modulation

of the ψ envelope enhances conversion efficiency and allows bidirectional transfer—charging or discharge via controlled ψ-gradients.

Representative Claims

1. An energy-harvesting apparatus comprising a resonant cavity or metamaterial lattice configured to sustain scalar refractive field (ψ) oscillations and an energy-conversion interface coupled to said lattice for transducing refractive potential variations into electrical power.
2. A method of storing energy comprising generating a spatial ψ gradient within a storage medium and maintaining said gradient to confine energy as a standing refractive potential.
3. A bidirectional converter capable of transforming electrical or mechanical input into modulated ψ energy and vice versa, the transformation efficiency governed by local $\nabla\psi$ amplitude.
4. A device wherein a network of superconducting or ferroelectric cells forms a distributed ψ resonator that accumulates energy proportional to $\int |\nabla\psi|^2 \, \mathrm{d}V$.
5. A system integrating ψ energy storage with electromagnetic induction modules to provide hybrid energy output and stabilization against external field perturbations.

Brief Description of the Drawings

Fig. 1 – ψ-field energy converter concept.
Fig. 2 – Refractive potential gradient to electrical output.
Fig. 3 – Bidirectional storage and release loop.
Fig. 4 – Scalable array of metamaterial energy cells.

Detailed Description

1. ψ-Field Converter Concept

Energy is extracted where $\nabla\psi \neq 0$, the local refractive potential storing usable density. A metamaterial cavity maintains standing ψ-waves whose nodes and antinodes correspond to alternating high- and low-energy regions.

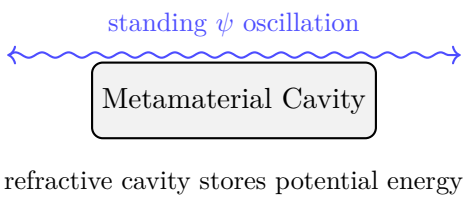

Figure 1: Metamaterial cavity sustaining oscillations of the ψ, converting refractive energy to electrical or mechanical output.

2. Gradient to Electrical Conversion

Local variation of ψ induces displacement currents in embedded conductive paths; rectified outputs yield electrical power.

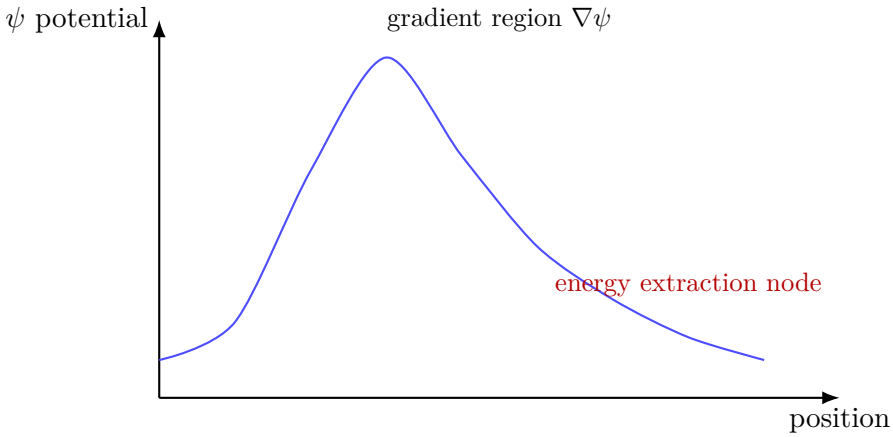

Figure 2: Conversion of refractive potential gradient into electrical power through embedded conductive interfaces.

3. Bidirectional Storage and Release

Energy can be stored by generating a controlled ψ-gradient and later released by reversing the modulation sequence.

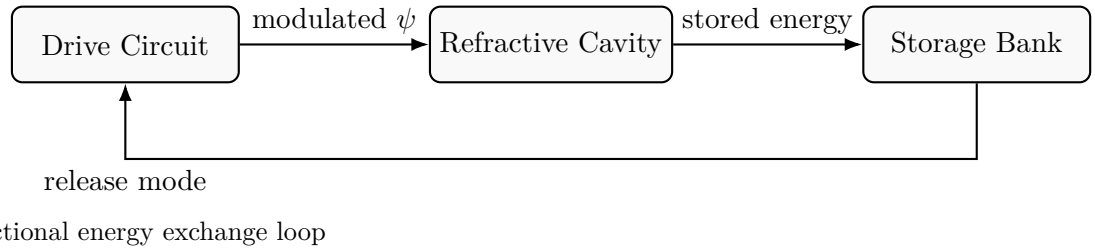

Figure 3: Closed-loop storage and release circuit coupling electronic drive and ψ resonator.

4. Scalable Array of Metamaterial Cells

Arrays of small ψ-resonators operate in phase to form large-scale power modules with minimal loss.

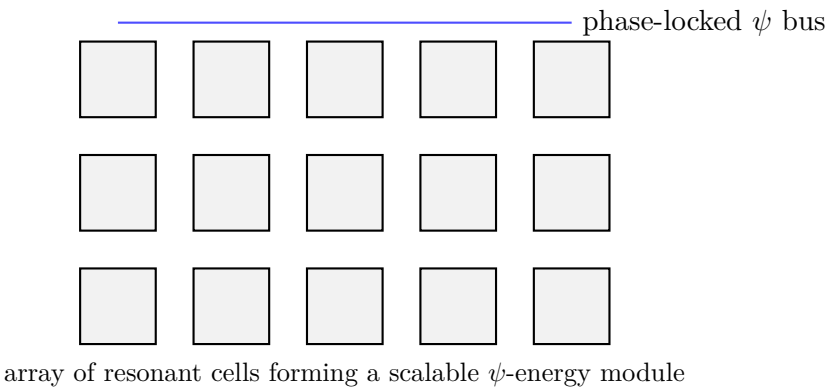

Figure 4: Modular array of ψ-resonant metamaterial cells with shared phase bus for scalable energy output.

Alternative Embodiments

(i) Hybrid photovoltaic–ψ converters combining sunlight excitation with refractive amplification. (ii) Oceanic or atmospheric ψ-harvesters exploiting ambient density gradients. (iii) Superconducting ψ-batteries storing refractive potential energy as persistent field configurations. (iv) Distributed ψ energy grids balancing local field intensity and load demand. (v) Compact ψ-capacitors for spacecraft or autonomous systems.

Industrial Applicability

Applicable to renewable energy generation, aerospace power systems, grid storage, portable electronics, and off-world energy harvesting where electromagnetic induction is inefficient.

Conclusion

By converting controlled refractive field gradients into usable electrical or mechanical energy, ψ-field energy harvesting and storage systems provide a new class of high-efficiency, loss-free power technologies. The architecture leverages metamaterial coupling and refractive-geometry storage to create sustainable energy solutions beyond electromagnetic limitations.